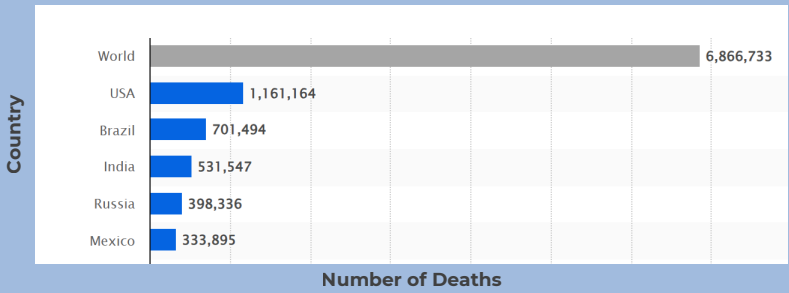


SARS-Covid Membrane Protein 7Y9B

Names: Larissa Nogueira Gomes, Jenna Desiderio, Lucas Ramondo

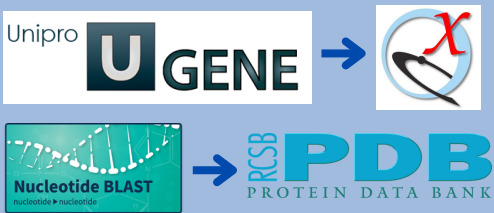
BACKGROUND

- As of May 2, 2023, **6 million people** have died from COVID-19. More needs to be done to address this issue.

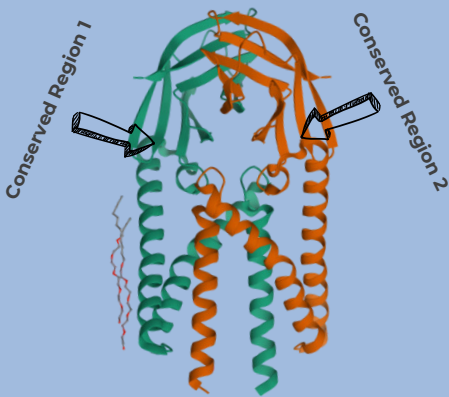


- To combat this, we are **investigating** the biological processes that undermine the characteristics of COVID.
- The **membrane (M) protein** is the most abundant structural protein of Covid and plays a central role in **virus assembly** via its interactions with various partner proteins (Wang et al, 2023).

INVESTIGATION PATHWAY



PROTEIN LOCATED: 7Y9B

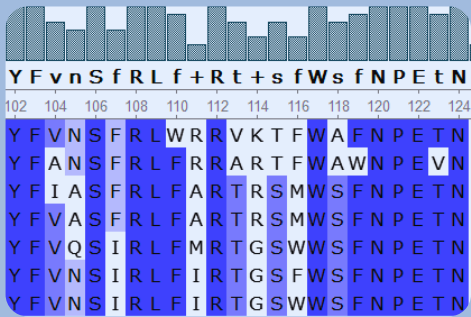


DRUG TARGET

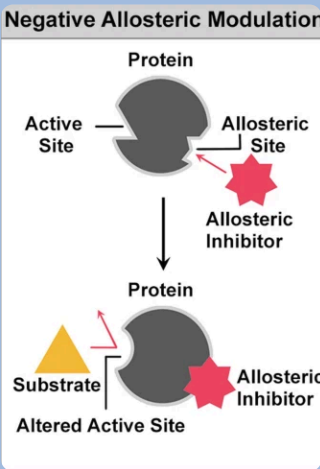
Protein	Amino Sequence	E-value
Human coronavirus OC43, complete genome	SLGIILLFITIIL	0.69
Human coronavirus HKU1 genotype B, complete genome	YFVNSIRLFIRTGSWWSFNPETNNL	0.034
Human coronavirus 229E strain 229E/Haiti-1/2016, complete genome	YFANSFRLFRRARTFWAWNPEVN	1.4

After investigating the conserved regions, KHU1 was chosen due to its low E-value. This sequence was then seen to be related to the **7Y9B** protein after insertion in the PDB

- A segment (**YFV-S- - L- -R-TSMWSFNPE**), located between **Codons 101** and **121**, is identified as **partially overlapping** with the interior membrane boundary of TMIII
- This is the **most conserved** (consensus 70-100%) motif in the M protein among all coronaviruses.
- This consensus could be related to **interaction with the N protein** (nucleocapsid) and viralTRNA



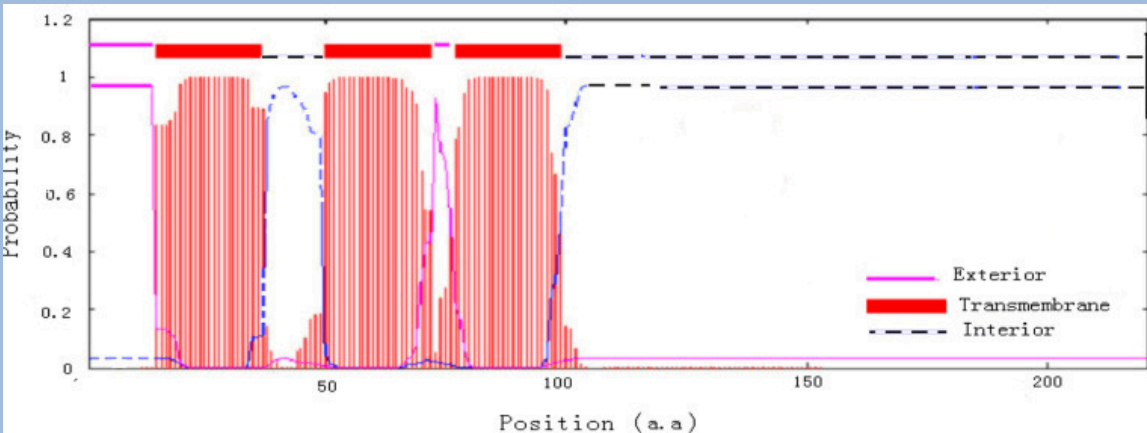
DRUG MOLECULE: ALLOSTERIC MODULATOR



- Binding Sites:** Bind to the TMIII Domain of the protein
- Orthosteric Antagonist:** Binds to the site and blocks the ligand and interactions from occurring
- This disrupts the activation of the receptor to **stop the interaction** with the N protein

From "Allostery in Drug Development," Xi Cheng & Hualiang Jiang, 2019.

SPECIFIC TARGET: TRANSMEMBRANE DOMAIN



- Region within membrane proteins, particularly G protein-coupled receptors (GPCRs) and ion transporters.
- It plays a crucial role in a protein's function, often serving as a key site for **ligand binding**, signal **transduction**, and **conformational changes** required for activation or channel gating.

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